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PETRONAS IN VIEW: The Functions of Management

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1.0 EXECUTIVE SUMMARY

In Malaysia, the national oil and gas corporation, well-known as PETRONAS, stands out in responsible production of energy and sustainable development. With more than 100 countries and a staff of around 50,000, PETRONAS has made its mark in the energy market. The corporation manages all areas of the oil and gas value chain, including midstream operations, downstream activities, and upstream exploration. In this report, we have chosen PETRONAS as a case study, and we have done a thorough analysis of it. The concise summary of the key findings and recommendations of the analysis is mentioned below.

1.1 Management Functions:

Management of PETRONAS incorporates planning, organization, leadership development, and control.

1. **Planning:** The long-term strategy of PETRONAS is to become a leading global energy organization. Besides, it also aims for net zero carbon emissions by 2050 and to inculcate sustainability into the core business operations. Tactical and operational approaches support this strategic view that helps extend market access and enhances operational efficiency.
2. **Organizing:** It fosters better decision-making and proper use of resources as a result of the hierarchical structure of PETRONAS. Four major areas that the organization encompasses comprise corporate functions, gas and new energy, downstream activities, and upstream operations. This design fosters good coordination among departments due to well spelled-out roles and responsibilities.
3. **Leadership Development:** PETRONAS believes that leadership is a source of competitive advantage, so it invests in the leaders through Universiti Teknologi PETRONAS and the PETRONAS Leadership Centre. The efforts undertaken to achieve strategic decision-making, innovation, and teamwork in building up a talent pool of competent executives who share common objectives of the company.
4. **Controlling:** PETRONAS has in place an effective control system that utilizes financial measures and KPIs in monitoring performance to ensure alignment with strategic objectives. It allows for adequate resource optimization, risk management, and continuous development in all aspects of its operations.

1.2 Challenges and Innovative Solutions:

Different issues related to PETRONAS include **migration towards renewable energy, operational inefficiency, digital transformation, and growing global competition.** Business requires innovative solutions for solving these challenges. Some unique solutions on which the business relies for solving problems include:

1. Creating hybrid energy centers by integrating renewable energy sources with the existing infrastructure and
2. Using AI-powered predictive maintenance solutions to improve operational efficiency by reducing chances of downtime.
3. Integration of blockchain technology into all business processes to enhance security and transparency.
4. Introduction of circular economy activities for increased sustainability and ethics in resource management.

2.0. COMPANY BACKGROUND

2.1. History

On October 17, 1974, PETRONAS was established as the national oil and gas company of Malaysia. Faced with the task of managing the nation's hydrocarbon resources, PETRONAS rapidly expanded its international presence in the 1980s and 1990s, moving into petrochemicals and gas processing (Fred et al., 2007). The company currently operates in more than 50 countries and is committed to sustainability, with investments in cleaner energy technologies, while remaining one of the major players in the energy sector worldwide (*PETRONAS (Petroliam Nasional)*, 2025).

2.2. Global Presence and Size

Working in over 100 nations, PETRONAS is recognized among the world's biggest enterprises by income, as proven by its positioning within the Fortune Worldwide 500®. The company employs nearly 50,000 people worldwide, underscoring its broad operational footprint and impact within the worldwide vitality segment (*Global Directory | PETRONAS Global*, 2025).

2.3. Industry and Operations

PETRONAS's operations span the complete oil and gas esteem chain, including:

1. **Upstream Investigation and Production:** The company is involved in the investigation and advancement of hydrocarbon assets, overseeing oil and gas generation resources both locally and globally (*About Us | PETRONAS Global, 2025*).
2. **Midstream Operations:** This portion incorporates gas preparation, liquefaction, and the operation of gas transmission pipelines, encouraging the effective transport and promotion of melted characteristic gas (LNG) universally.
3. **Downstream Exercises:** PETRONAS is included in refining rough oil, showcasing and dispersing petroleum items, fabricating petrochemicals, and overseeing shipping coordinations, in this manner guaranteeing a consistent supply chain from generation to end-user conveyance.

2.4. Key Aspects of Business Operations

1. **Coordinates Trade Show:** PETRONAS's comprehensive integration across the oil and gas value chain empowers it to capture synergies and optimize efficiencies, improving its competitiveness within the worldwide energy market.
2. **Supportability Commitment:** Illustrating a commitment to capable vitality generation, PETRONAS centers on conveying vitality arrangements that progress societal advancement while following feasible hones.
3. **Operational Execution:** The company emphasizes persistent enhancement in operational execution through advanced activities, accomplishing noteworthy gains in preparing security, unwavering quality, and fetched proficiency to meet broader corporate goals.

Obviously, PETRONAS's advancement from a national substance to a worldwide energy powerhouse reflects its key integration with the oil and gas industry, commitment to supportability, and persistent interest in operational excellence.

3.0. FUNCTIONS OF MANAGEMENT

3.1. Planning at PETRONAS:

At Petroliam Nasional Berhad (PETRONAS), planning is integral to navigating the complexities of the oil and gas industry, especially in the context of evolving market

dynamics and sustainability commitments. Therefore, PETRONAS employs a comprehensive strategic planning framework that encompasses various levels of planning:

1. **Corporate Strategic Planning:** This involves long-term goals and directions aligned with PETRONAS's vision of becoming a leading global energy player. The strategic plan focuses on integrating sustainability into core operations while ensuring profitability and growth. For instance, PETRONAS has committed to achieving net zero carbon emissions by 2050, which is a significant aspect of its strategic planning efforts (PETRONAS, 2022a).
2. **Tactical Planning:** Tactical plans are developed to translate strategic goals into actionable steps. These plans typically cover shorter time frames (1-3 years) and include specific initiatives such as enhancing operational efficiencies, expanding international markets, and investing in new technologies (PETRONAS, 2022b). For example, according to *PETRONAS Accelerates Energy Transition Strategy with a Diverse Portfolio to Drive Future Growth | PETRONAS Global* (2024a), PETRONAS has outlined its intention to intensify domestic oil and gas exploration while pursuing low-carbon energy solutions.
3. **Operational Planning:** This level focuses on day-to-day operations and includes detailed action plans that guide employees in executing their tasks efficiently. Operational plans are essential for ensuring that tactical objectives are met through effective resource allocation and management.

3.1.1. Significance of Planning at PETRONAS

1. **Alignment with Vision and Mission:** Planning at PETRONAS ensures that all organizational activities align with its overarching vision and mission. This alignment helps in maintaining focus on long-term objectives while adapting to short-term challenges.
2. **Risk Management:** The planning process incorporates risk assessment strategies to identify potential challenges and develop mitigation plans. For instance, during the COVID-19 pandemic, PETRONAS activated its Pandemic Preparedness and Response Team to ensure operational continuity while safeguarding employee health (*Moving Forward with Passion Strategic Progress*, n.d.).

3. **Resource Optimization:** Effective planning allows PETRONAS to allocate resources efficiently across various projects and divisions. This optimization is crucial in an industry characterized by high capital expenditures and fluctuating market conditions.
4. **Stakeholder Engagement:** Through strategic planning, PETRONAS engages with stakeholders—including investors, regulators, and communities—to ensure transparency and alignment with societal expectations. This engagement is vital for building trust and securing support for initiatives such as environmental sustainability efforts (*PETRONAS Accelerates Energy Transition Strategy with a Diverse Portfolio to Drive Future Growth* | PETRONAS Global, 2024b).
5. **Innovation and Adaptation:** The dynamic nature of the energy sector necessitates continuous innovation. PETRONAS's planning processes emphasize research and development, enabling the company to adapt to technological advancements and changing consumer preferences for cleaner energy solutions (PETRONAS, 2022c).

3.1.2. Impact of Planning on PETRONAS's Organizational Performance

1. **Enhanced Resilience:** Based on the findings in (*PETRONAS Accelerates Energy Transition Strategy with a Diverse Portfolio to Drive Future Growth* | PETRONAS Global, 2024c), the proactive approach to planning has enabled PETRONAS to navigate economic downturns and industry disruptions effectively. By implementing measures like cost rationalization during challenging times, the company preserves value while maintaining operational integrity.
2. **Sustainable Growth:** Strategic initiatives focused on sustainability not only enhance PETRONAS's reputation but also position it for long-term growth in a transitioning energy landscape. The commitment to reducing carbon emissions aligns with global trends toward renewable energy sources.
3. **Improved Employee Engagement:** Clear planning processes foster a sense of purpose among employees by aligning their roles with organizational goals. This engagement is crucial for driving performance and innovation within the workforce.

3.2. Organizing at PETRONAS

Organizing encompasses the establishment of a structured framework that facilitates effective decision-making, resource allocation, and coordination among various business units. This

function is essential for managing the complexities of a multinational corporation like PETRONAS operating in the oil and gas sector. Hence, PETRONAS adopts a hierarchical organizational structure that allows for clear lines of authority and responsibility. The company is divided into several key divisions, each focusing on specific aspects of its operations:

1. **Upstream Operations:** This division is responsible for exploration and production activities, including oil and gas extraction.
2. **Downstream Operations:** Focused on refining, distribution, and marketing of petroleum products, this division ensures that products reach consumers effectively.
3. **Gas and New Energy:** This segment emphasizes natural gas operations and the transition to renewable energy sources.
4. **Corporate Functions:** Including finance, human resources, legal, and strategic communications, these functions support the core business operations.

Based on the information extracted from *How We Are Structured | PETRONAS Gas Berhad (PGB) (2023)*, *PETRONAS Pengerang Integrated Complex (PIC) (2025)*, as well as *PETRONAS | a Leading Oil and Gas Multinational of Choice (2025)*, the organizational chart illustrates the various leadership roles within PETRONAS, highlighting key positions such as:

- Chief Executive Officer (CEO): Oversees overall strategic direction.
- Senior Vice Presidents: Manage specific divisions or functions.
- Department Heads: Responsible for day-to-day operations within their respective areas.

3.2.1. Significance of Organizing at PETRONAS

1. **Clarity in Roles and Responsibilities:** The structured hierarchy at PETRONAS ensures that employees understand their roles and how they contribute to the organization's objectives. This clarity enhances accountability and performance across all levels.
2. **Efficient Resource Allocation:** By organizing resources effectively, PETRONAS can optimize its operational efficiency. Each division is equipped with the necessary tools and personnel to perform its functions effectively, reducing waste and improving productivity.

3. **Facilitation of Communication:** The organizational structure fosters communication both vertically (between different levels of management) and horizontally (across departments). This is crucial for collaboration on projects that require input from multiple areas of expertise.
4. **Adaptability to Change:** PETRONAS's organizing function enables it to respond swiftly to changes in the market or regulatory environment. The clear structure allows for quick decision-making processes, which are essential in the dynamic energy sector.
5. **Focus on Strategic Goals:** The organization aligns its structure with strategic goals such as sustainability and innovation. For example, dedicated teams work on carbon capture technologies and renewable energy initiatives, ensuring that these priorities are integrated into daily operations (*How We Are Structured* | PETRONAS Gas Berhad (PGB), 2023b) (PETRONAS Pengerang Integrated Complex (PIC), 2025b).

3.2.2. Impact of Organizing on PETRONAS's Organizational Performance

The impact of effective organizing at PETRONAS manifests in several ways:

1. **Enhanced Operational Efficiency:** A well-organized structure leads to streamlined processes, reducing delays in project execution and improving overall operational performance.
2. **Stronger Competitive Position:** By optimizing its organizational framework, PETRONAS can respond more effectively to competitive pressures in the global energy market.
3. **Improved Employee Morale:** Clarity in roles and responsibilities contributes to higher employee satisfaction, as individuals feel more engaged and valued within a structured environment.
4. **Sustainability Commitment:** The organization's focus on integrating sustainability into its operations reflects its commitment to responsible resource management while meeting stakeholder expectations.

3.3. Leadership Development at PETRONAS

PETRONAS views leadership as a competitive advantage. It produces leaders who can handle difficult problems and promote sustainable growth through well-run organizations and programs (PETRONAS, 2022b).

3.3.1. Key Institutions:

1. **PETRONAS Leadership Centre (PLC):** PLC, which was founded in 1979, is the hub for leadership development. Programs catered to various management levels, such as "Leaders Developing Leaders," where top executives coach up-and-coming talent, are part of its creative approach (PETRONAS, 2022b). This ensures congruence with business objectives by fostering a culture of cooperation and knowledge exchange. The influence of PLC is demonstrated by its capacity to develop leaders who are exceptional at strategic decision-making and innovation, two skills that are essential to preserving PETRONAS's competitive advantage (Zawawi et al., 2022).
2. **Universiti Teknologi PETRONAS (UTP):** UTP combines leadership development with academic achievement, emphasising sustainability, engineering, and technology. UTP equips students to assume leadership positions in the energy industry by integrating leadership concepts into its curriculum (Iftekhhar, 2022). This guarantees a consistent flow of qualified experts prepared to support PETRONAS's long-term goals. Because it supports PETRONAS's efforts in technical developments, the university's emphasis on research and innovation further increases its relevance.
3. **Institut Teknologi Petroleum PETRONAS (INSTEP):** INSTEP uses real-world, hands-on methods to deliver technical and leadership training. Its competency-based programs guarantee that staff members possess leadership and problem-solving abilities in addition to technical proficiency (PETRONAS, 2022b). In a highly competitive business, PETRONAS's operational effectiveness and ongoing resilience are strengthened by this dual focus.
4. **SEEd.Lab:** SEEd.Lab develops entrepreneurial character while using technology to address social issues. This effort is noteworthy because it creates leaders who are both socially concerned and business-savvy, thereby coordinating PETRONAS's CSR with its leadership objectives.

5. **Programs and Initiatives Managerial Excellence (ME) Program:** This program, which is designed for mid-level managers, uses cutting-edge training technology to improve skills related to leadership (PETRONAS, 2022b). Participants' enhanced performance and decision-making skills, which have a direct impact on organizational success, demonstrate its influence.
6. **BeDigital Bootcamp:** This program bridges the gap between educational institutions and business by giving local graduates the fundamental digital skills they need. It ensures flexibility in the era of the technological revolution by fortifying PETRONAS's skill resources.
7. **Innovation Masterclasses:** These PLC-offered workshops help executives develop an innovative and creative attitude, which promotes a culture that propels profitable growth and ongoing enhancement (PETRONAS, 2022b).

Significance and Impact of PETRONAS's programs for developing leadership are essential to maintaining its standing as a world leader in energy. The corporation protects its organizational success and makes a substantial contribution to national development and industry standards by cultivating creativity, adaptability, and moral leadership (Zawawi et al., 2022).

3.4. Controlling at PETRONAS

3.4.1. How it Performs:

PETRONAS's robust control system ensures that the company meets its strategic objectives. This means continuously monitoring and evaluating performance at various organizational levels via Key Performance Indicators (KPIs) along with additional performance measurements (PETRONAS, 2022a). The organization utilizes both financial and non-financial controls to measure the effectiveness of its operations in order to ensure that resources are employed efficiently and that activities are in line with its overall strategy.

3.4.2. Performance Metrics:

To assess how well various departments and units are performing, PETRONAS uses key performance indicators (KPIs) including revenue growth, profits, cost control, and productivity (PETRONAS, 2022a). The company's shifting requirements and its outside environment are reflected in the regular evaluation and adjustment of these indicators.

3.4.3. Budgeting and Financial Control:

A key component of PETRONAS's operations is financial control. The business makes sure that its resources are distributed efficiently and that it maintains its financial stability through budgeting and financial projections (Iftekhhar, 2022). Any budgetary gaps are closely watched, and where needed, correction is taken.

3.4.4. Operational Control:

For increased performance in daily operations, PETRONAS also employs operational controls. This covers scheduling, allocating resources, and maintaining standards of quality. To determine if operations achieve the intended goals, audits and performance reviews are carried out on a regular basis (PETRONAS, 2022a).

3.4.5. Significance and Impact: Ensuring Alignment with Strategic Goals:

At PETRONAS, the controlling function is essential to making sure that all operations are in line with the overarching strategic objectives of the business. PETRONAS can promptly detect any disconnects between projected and actual outcomes by tracking performance using KPIs and financial measurements. This enables prompt remedial measures to maintain the organization's direction.

3.4.6. Resource Optimization:

The regulating system makes sure that human and financial resources are used as efficiently as possible. Regular budget reviews and performance evaluations allow PETRONAS to spot inefficiencies and reallocate resources as needed (Iftekhhar, 2022). This is especially crucial in resource-intensive industries like energy, where success depends on controlling expenses and optimizing output.

3.4.7. Risk Management:

Controlling also plays a critical role in risk management. Through consistent performance monitoring against predetermined benchmarks, PETRONAS can detect possible risks—financial, operational, or strategic—early (PETRONAS, 2022a). By taking a proactive stance, risks are reduced before they become more serious, guaranteeing long-term growth and company endurance.

3.4.8. Continuous Improvement and Adaptation:

PETRONAS's attempts to continuously develop are supported by its controlling function. Through regular performance monitoring, the business may pinpoint opportunities for development and best practices (Iftekhhar, 2022), promoting an innovative and efficient

culture. In an industry like energy that is competitive and changing quickly, this flexibility is crucial.

3.4.9 Impact on Decision Making:

Senior management receives the information and insights required for well-informed decision-making from the controlling system. The managing role makes sure that decisions are founded on precise, current performance data (PETRONAS, 2022a), which produces more effective results, whether it is modifying operational strategies or investigating potential markets.

4.0. PROBLEM ANALYSIS AND INNOVATIVE SOLUTION:

KUALA LUMPUR in (Aug 28): The national oil company Petroliam Nasional Bhd (Petronas) is in a hard place. Other than the regular issues brought about by the Covid-19 pandemic, such as destruction to business activities, which in turn have slowed down the demand for oil, the company finds itself caught between having to reinvest into cleaner sources of fuel to ensure the longevity of its business while at the same time monetizing existing oil and gas assets and looking at possibly hiving off some of these producing assets. As the national oil company, Petronas serves as the custodian of the country's petroleum resources. So, with the more than 90% gain of oil prices since end-October last year, the national oil company is very much expected to pay its shareholder—the federal government—a higher quantum of dividends in order to help make both ends meet in the challenging climate today (*The Many Challenges Facing Petronas*, n.d.).

4.1. Problem Analysis:

PETRONAS, among the largest oil and gas companies in the world, experiences problems that are common in this industry. The major concerns include:

1. **Transition to Renewable Energy:** Because of the world's fight against climate change, PETRONAS faces pressure to transition into renewable energy sources from a dependency on fossil fuels (Gielen et al., 2019). Transitioning into renewable energy will have huge investments in costs, technologies, and re-skilling the workforce.

2. **Operational Efficiency:** The major risks to profitability include ageing infrastructure, a volatile oil price, and operational inefficiency. Delays in regular maintenance and unplanned shutdowns further aggravate the problem.
3. **Digital Transformation:** While a digital strategy has been initiated by PETRONAS, its full implementation across all operations is still an uphill task. Data silos, cybersecurity concerns, and resistance to change among employees contribute to hindering progress.
4. **Sustainability Goals:** To reach net-zero carbon emissions by 2050, CCUS and greener ways of production urgently need new strategies.
5. **Global Competition:** In light of volatile geopolitical landscapes, it is high time for innovative and cost-effective solutions to enable PETRONAS to maintain its competitive position among other major energy players.

4.2. Innovative Solutions for PETRONAS Challenges

1. Transition to Renewable Energy:

Solution: Hybrid Energy Hubs

- **Renewable-Integrated Facilities:** Establish hybrid energy hubs that integrate solar, wind, and geothermal energy with existing fossil fuel infrastructure. These hubs would use advanced battery storage to store surplus renewable energy or convert it into green hydrogen for industrial and residential use.
- **Renewable Energy Zones:** Identify optimal locations in Malaysia with high renewable energy potential, such as coastal areas suitable for offshore wind farms or regions with ample sunlight for solar power. Collaborating with global experts can enable the development of large-scale renewable energy projects, helping PETRONAS meet its net-zero goals.

2. Operational Efficiency

Solution: Advanced Predictive Analytics and Digital Twin Technology

- **AI-Powered Predictive Maintenance:** Implement AI-driven systems that use real-time data from IoT sensors to predict equipment failures, optimize maintenance schedules, and enhance the longevity of assets. Predictive analytics will reduce maintenance costs and downtime, increasing the operational efficiency of existing infrastructure.

- **Digital Twin Technology:** Create virtual replicas of physical operations to simulate, monitor, and optimize real-time processes. This technology will enable PETRONAS to test new scenarios, identify inefficiencies, and optimize production processes without disrupting daily operations.
- **Mobile Workforce Platforms:** Introduce AI-powered mobile platforms for field employees, offering real-time diagnostics, instructions, and decision-making support, empowering the workforce to operate efficiently and safely.

3. Digital Transformation:

Solution: Blockchain for Transparency and Cybersecurity

- **Unified Blockchain System:** Utilize blockchain technology to create a unified, transparent supply chain management system that ensures the security and accuracy of transactions, contracts, and payments. Blockchain can reduce operational costs, eliminate data silos, and enhance collaboration across various business units.
- **Cybersecurity Enhancement:** With the rise of digital transformation, cybersecurity is crucial. PETRONAS should partner with top cybersecurity firms to develop cutting-edge security systems and conduct regular training for employees to mitigate the risks of cyberattacks and data breaches.

4. Achieving Sustainability Goals:

Solution: Circular Economy and Carbon Offset Initiatives

- **Carbon Offset Marketplace:** PETRONAS could launch a carbon offset platform that enables businesses and consumers to purchase carbon credits generated through the company's renewable energy projects, thus promoting its sustainability efforts and contributing to global emission reduction targets.
- **Biomass Utilization:** Invest in converting agricultural and industrial waste into biofuels and bioplastics. These initiatives will help PETRONAS reduce waste and contribute to the circular economy, where production is more sustainable and resource-efficient.

5. Addressing Global Competition:

Solution: Collaborative Innovation and Customer Co-Creation

- **Global R&D Alliances:** Form strategic research and development partnerships with international institutions and tech companies to co-develop low-carbon fuel technologies, cost-effective renewable energy solutions, and cutting-edge industry

technologies. This collaboration will accelerate innovation and strengthen PETRONAS's competitive edge.

- **Customer Co-Creation Platforms:** Build platforms that involve customers in the product innovation process, enabling them to suggest and co-design sustainable solutions. This will help PETRONAS better align its offerings with customer demands while fostering a sense of community and shared responsibility.

4.3. Expected Impact:

By implementing such strategies, PETRONAS will be able to achieve operational efficiency and cost reduction, transition smoothly into renewable energy while achieving its net-zero goals, and strengthen its market position as a sustainable energy leader, fostering innovation and collaboration across its ecosystem.

5.0. CONCLUSION

In conclusion, PETRONAS has withstood the increasingly complex web of challenges. Through strategic and organizational focus, an ongoing process of leadership development, and good control, the group is performing exceptionally on most parameters of each segment of its operations. The company has set itself on a road to sustainability through plans like net zero carbon emissions by 2050 and is thus well on the trail into renewable transitions.

However, several big challenges still lie ahead for PETRONAS: the shift to renewable energy from fossil fuels, operational efficiency, and a complete digital transformation. Against such a backdrop, innovative solutions lie in placing PETRONAS in the best position to address hybrid energy hubs, AI-powered predictive maintenance, and blockchain integration. These moves not only smoothen existing operations but also set up long-term sustainability and growth in the environment of energy transition.

With a commitment to the responsible production of energy added to driving innovation, PETRONAS is uniquely positioned to play a leading role in shaping the future of the global energy industry. The efforts it invests in the alignment of business operations with sustainable development goals will ensure benefits not only for the company but also for global efforts toward combating climate change and ensuring a more sustainable future for all.

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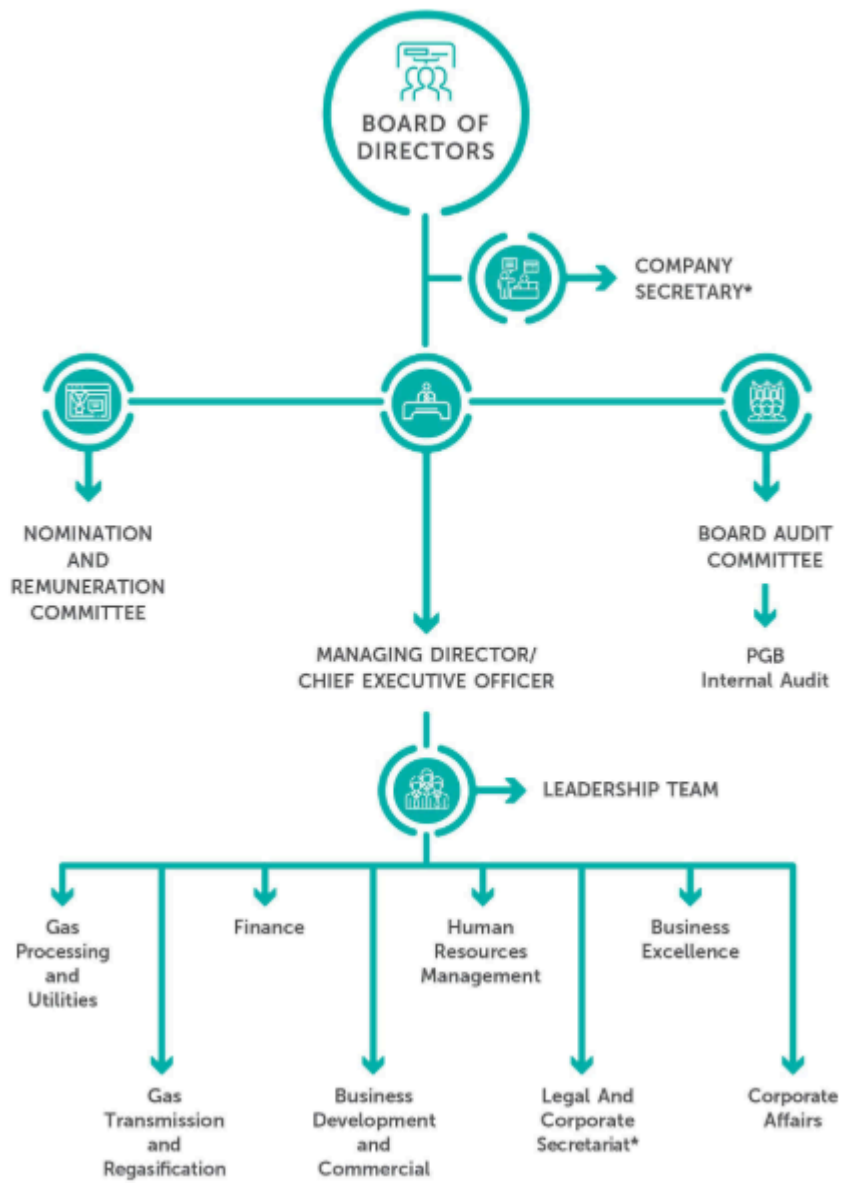
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APPENDICES

PGB Organisation Design & Structure



Company Structure

